## Q1.

Some pupils predicted that water will evaporate faster if the surrounding air temperature is higher.

To investigate their prediction they placed some water in containers in two different rooms.

(a) Give **two** factors they should keep the same to make their investigation fair.

1. .....

1 mark

2. .....

1 mark

(b) They recorded the mass of the water and the container in room 1 and room 2 every day for 5 days.

The table below shows their results.

| time (days) | mass of water and container (g) | |
|---|---|---|
| | room 1 | room 2 |
| 0 | 100 | 100 |
| 1 | 92 | 85 |
| 2 | 80 | 72 |
| 3 | 72 | 54 |
| 4 | 60 | 45 |
| 5 | 46 | 30 |

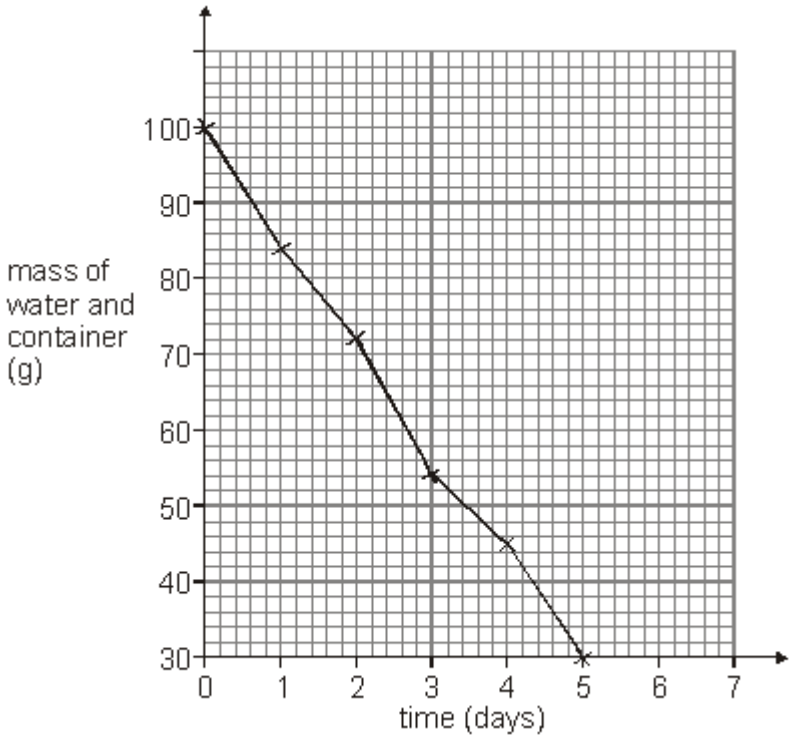
The data shown in their table is **not** sufficient to test their prediction.
Explain why.

.....

.....

1 mark

They plotted their data for room 2 and attempted to draw a line of best fit.




(c) Describe the mistake they made in drawing the line of best fit.

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1 mark

(d) Using the data in the table plot the points for room 1.

1 mark

(e) Draw a line of best fit of the points you have drawn.

1 mark

(f) In which room did the water evaporate more quickly?
Tick **one** box.

room 1 ☐ room 2 ☐

Use their data to explain your answer.

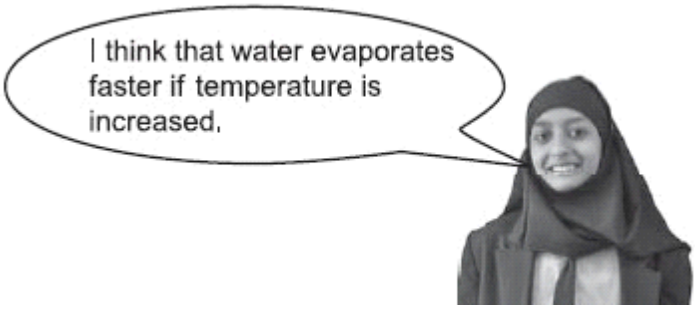
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1 mark
maximum 7 marks

**Q2.**

Amena described her idea about the evaporation of water.




Amena

(a) Write a plan for an investigation you could carry out in the school laboratory to test Amena's idea.
Assume you have access to all the usual laboratory equipment.

In your plan you must write:

- the one factor you would change as you carry out your investigation (the independent variable)
- the effect you would observe or measure as you carry out your investigation (the dependent variable)
- one factor you would keep the same to help make your test fair.

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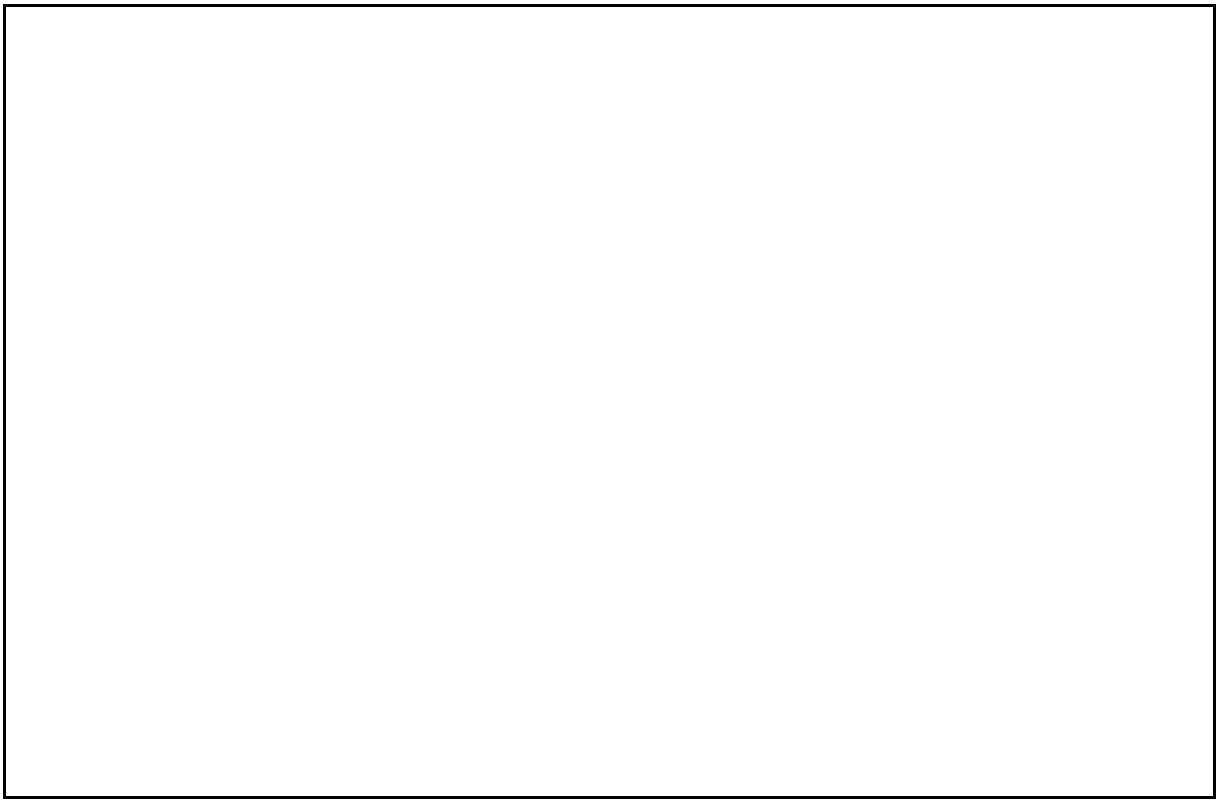
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3 marks

(b) In the box below, draw and label a table that you could use to record your results.



1 mark
maximum 4 marks

## Q1.

(a) any **two** from

- initial temperature of water
- exclusion of draughts
  *accept 'same amount of draughts'*
- volume **or** mass of water
  *accept 'amount of water'*
- surface area (of water)
  *accept 'shape **or** size **or** material of container'*
- humidity
  *accept 'the time it is left for'*
  *'temperature' is insufficient*
  *do not accept 'size of room'*

**2 (L5)**

(b)
- they have not collected data about temperature
  *accept 'they did not measure temperature'*
- no data recorded about temperature
  *accept 'no information about temperature'*
  *accept 'no temperature'*

**1 (L5)**

(c) the line is not straight (converse)
the line should be straight
*accept 'the line is zig-zag'*
*accept 'they joined up all the points'*
*accept 'they didn't go between the points'*
*accept 'the line does not need to go through all the points'*

**1 (L6)**

(d) 6 points correctly plotted to the nearest half square

**1 (L6)**

(e) a straight line of best fit between some of the points

**1 (L6)**

(f)

☐ ☑ room 2

and

- the points for room 2 are lower than room 1
- the container in room 2 reaches 30g in 5 days, room 1 reaches 45g
  *accept 'there is less water in the container in room 2 than in*

*room 1'*

- more water evaporated in the same time

  *accept 'amount of water dropped more in the first three days'*
  *accept 'the line for room 2 is steeper'*

- the volume goes down more quickly

  *accept 'the numbers go down more quickly'*
  *accept 'the mass goes down more quickly'*

**1 (L6)**

**[7]**

## Q2.

**Markers should read the answers to parts a and b before marking this question.**

**Parts a and b should be marked together.**

(a) • temperature of the water

*accept 'temperature'*
*accept 'room temperature'*
*do **not** accept responses that describe rates of heating.*

**1 (L7)**

any **one** from

- rate of evaporation

  *accept 'the time for it to evaporate'*
  *answers must refer to both time taken and amount of water lost*

- time taken for all the water to evaporate

  *accept 'measure how much water is left after a certain time'*
  *'time taken' is insufficient*

- volume **or** mass **or** amount of water lost in a fixed time

**1 (L7)**

any **one** from

- starting volume of water

  *accept 'the amount of water'*
  *accept a specified volume of water*
  *'same heater' and 'same starting measurement' are insufficient*

- shape of container

- same ambient conditions

  *accept 'room temperature' if the independent variable is 'water temperature'*

**1 (L7)**

(b) a column **or** row indicating temperature **and** a column **or** row indicating time **or** volume lost **or** volume remaining

*accept a column **or** row indicating 'rate of evaporation'*
*accept 'amount lost' **or** 'amount remaining'*

***both** headings are required for the mark*
*the units of measurement are not necessary for the mark*
*the second column **or** row should be consistent with the dependent variable identified in part **a***
*ignore other columns in the table*

**1 (L7)**

**[4]**